

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – TRANSFORMER MECHANICS WORKSHEET

Course Code:	CSA6503	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 3 – Transformer Mechanics Worksheet
Weightage:	40% of CIA	Total Marks:	100 (1 Question)
CO1 Statement:	<i>Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models. (BL1, BL2)</i>		
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Section / Batch:	SlotA/2024	Date:	27/07/2026

Instructions to Students

- Answer one questions. It carries 100 Marks. Total: 100 Marks.
- Tokenize a sentence.
- Convert tokens to token IDs.
- Explain embeddings.
- Label a Transformer encoder.
- Identify Query, Key, and Value.
- Explain which words receive higher attention.
- Compare Transformer models.
- Complete a Transformer pipeline.
- Draw a block diagram.

Code of Conduct

I ch.poorna teja(192472149) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Transformer Decoder Components Worksheet

Introduction

The Transformer Decoder is one of the core components of the Transformer architecture introduced in the paper "Attention Is All You Need". It is primarily designed for sequence generation tasks, where the model predicts the next word or token based on previously generated words. Unlike the encoder, which focuses on understanding the input, the decoder generates meaningful and coherent output one token at a time.

The decoder uses masked self-attention to ensure that future words are not visible while predicting the current word. It also uses encoder-decoder attention to focus on relevant parts of the input sequence produced by the encoder. These mechanisms enable transformer models to generate accurate translations, summaries, responses, and other forms of text.

Modern AI models such as GPT, T5, and BART use transformer decoder architecture for text generation. Decoder-based transformers are widely used in chatbots, virtual assistants, code generation, story writing, and machine translation because of their ability to generate fluent and context-aware text.

Understanding the components of the transformer decoder helps students and researchers learn how modern language models generate human-like responses. Each component performs a specific function, contributing to efficient sequence generation and contextual understanding.

Comparison of Popular Transformer Models

Component	Main Function	Key Features
Masked Multi-Head Self-Attention	Prevents access to future tokens	Uses causal masking for autoregressive generation
Encoder-Decoder Attention	Connects decoder with encoder output	Focuses on relevant input information
Feed Forward Neural Network (FFN)	Learns complex feature representations	Two fully connected layers with activation
Add & Layer Normalization	Stabilizes and speeds up training	Residual connection + Layer Normalization
Positional Encoding	Preserves word order	Adds positional information to token embeddings
Linear Layer & Softmax	Predicts the next token	Converts decoder output into a probability distribution

Important Comparison Points

1. Masked Multi-Head Self-Attention

- Allows each token to attend only to previous tokens.
- Prevents future information leakage.
- Uses multiple attention heads to learn different relationships.
- Essential for autoregressive text generation.

2. Encoder-Decoder Attention

- Receives output from the encoder.
- Helps the decoder focus on important input words.
- Improves translation and summarization quality.
- Connects input sequence with generated output.

3. Feed Forward Neural Network (FFN)

- Processes each token independently.
- Consists of two fully connected layers.
- Uses activation functions like ReLU or GELU.
- Learns higher-level feature representations.

4. Add & Layer Normalization

- Adds residual (skip) connections.
 - Applies Layer Normalization after each sub-layer.
 - Prevents vanishing gradients.
 - Improves training stability and convergence.
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5. Positional Encoding

- Adds position information to embeddings.
 - Helps the model understand word order.
 - Uses sinusoidal or learned positional embeddings.
 - Enables processing of sequences without recurrence.
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6. Linear Layer & Softmax

- Converts decoder outputs into vocabulary scores.
 - Softmax generates probability distribution.
 - Selects the most probable next token.
 - Produces the final generated word.
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Advantages

- Efficient parallel computation during training.
 - Generates fluent and context-aware text.
 - Handles long-range dependencies effectively.
 - Supports multiple NLP applications.
 - High scalability for large language models.
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Limitations

- High computational cost.
 - Requires large memory resources.
 - Needs massive datasets for training.
 - Decoder-only models cannot directly access future tokens.
 - Large models require powerful GPUs/TPUs.
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Applications of Transformer Decoder

- Chatbots and conversational AI
 - Text generation
 - Machine translation
 - Story and article writing
 - Code generation
 - Text summarization
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- Question answering
- Email and content generation

Conclusion

The Transformer Decoder is a powerful sequence generation component that enables modern AI systems to produce natural and meaningful text. Its masked self-attention mechanism ensures that predictions depend only on previously generated tokens, while encoder-decoder attention allows the model to utilize information from the input sequence effectively. Additional components such as the Feed Forward Network, Layer Normalization, Positional Encoding, and Softmax layer work together to improve accuracy, stability, and efficiency.

Despite requiring significant computational resources and memory, transformer decoders have become the foundation of state-of-the-art language models such as GPT, T5, and BART. They are widely used in applications including chatbots, machine translation, content creation, code generation, and text summarization. As research advances, transformer decoder architectures continue to become faster, more efficient, and more capable, driving the next generation of intelligent AI systems.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical / Concept	30%	All key concepts (AI, ML, DL, GenAI, LLMs) correctly identified with clear hierarchical relationships	Most concepts correct; minor gaps in relationships	Several concepts missing or misclassified	Major conceptual errors; map incomplete
Application	25%	Real-world examples linked accurately to each concept	Examples mostly relevant	Few or generic examples	No real-world linkage shown
Quality	20%	Logical, well-organized structure with clear connectors	Mostly organized; minor structural issues	Some disorganization affecting clarity	Poorly structured, hard to follow
Presentation	15%	Visually clear, creative, easy to interpret	Neat and readable	Cluttered or inconsistent formatting	Untidy, difficult to interpret
Documentation / Communication	10%	Concise labels/notes that add clarity	Adequate labeling	Minimal or unclear labeling	No supporting labels or explanation

Marks Evaluation:

Question No.	Technical / Concept(30%)	Application(25%)	Quality (20%)	Presentation (15%)	Documentation / Communication (10%)	Total Marks (100)
Q1						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____